Safe Smart Maps - UI Design Brief for Stitch

Cal Hacks 13.0 | Version 0.2 | 24 Oct 2026 | Source of truth for behavior: docs/specs/ and docs/agent-context/CONTRACTS.md. This brief is for the person or tool designing the screens. Numbers and place names shown are **sample content** for design only.

1. What we are building

Normal maps show the fastest walking route. Safe Smart Maps shows **three** walking routes for the time the user will actually be on the street: **Most lit & active** (slowest), **Balanced**, and **Fastest** (least lit and active). Each route is scored on open places, lighting, main roads, help points and transit, and, most importantly, on whether those things are spread along the **whole** route. A route with a busy first 20% and an empty last 80% must not look good. Colored segments on the map make weak stretches obvious.

2. Who it is for

Maya, 20, undergrad. Leaves the library at 10:40 PM and walks 15-25 minutes to transit or home, often alone, on her phone, one-handed, tired eyes, low battery. She will trade a few extra minutes for a route that stays lit and has open places along the whole way, and she wants to see *why* a route is recommended. Secondary viewers: hackathon judges watching a 2-minute demo on a laptop projector.

3. Design principles

- **Platform independent, mobile first:** one responsive design that works on Android phones and iPhones (375 x 812 and small 360 x 640), plus a Samsung Wear OS watch companion. One-handed, thumb-reachable controls at the bottom. Desktop is a bonus: panel on the left, map on the right.
- **Night-friendly:** default dark theme (dark map style), high contrast, large type, no pure-white glare. Light theme optional.
- **Calm, not scary.** No alarming imagery (shields, warning skulls, flashing red). We describe conditions; we do not promise safety.
- **Always show the reasons** and the time the data was fetched. Trust comes from transparency.
- **Color is never the only signal.** Green/yellow/red segments also carry an icon or line pattern and a text label (color-blind safe).
- **Honest about gaps:** missing or old data is shown as "not sure", never a guessed value. Activity is labeled "estimate".

4. Mandatory wording rules (copy on every screen)

Do	Don't
"Better lit and more active"	"Safe" as a promise, "secure", "dangerous", "crime-free"
Route labels: **Most lit & active**, **Balanced**, **Fastest**	Any claim that guarantees protection
"Not sure" when data is missing or stale	Filling blanks with guesses
"Activity estimate" (never a people count)	"12 people nearby"
"Unusual heart rate detected"	Medical claims or diagnosis
Footer: "Explained by AI, may be wrong. In an emergency call 911."	Hiding the AI/emergency note
Heart-rate readings in demos labeled "Simulated"	Showing simulated data as real

5. Visual system (starting point; Stitch may refine)

Item	Spec
Theme	Dark default. Background #0F1720, surface #17222E, text #F2F5F8, muted text #9FB0C0
Segment bands	Better lit/active (green) #2FBF71, Mixed (yellow) #F2B632, Weak (red-orange) #E5603D, Not sure (grey, dashed) #8A97A3. Each also gets an icon: check / dash / minus / question mark
Route identity	Most lit & active = teal accent, Balanced = blue, Fastest = violet (card accents only). Selected route is thick; others thin and semi-transparent
Typography	Clean sans-serif (Inter or similar). Title 24, card title 18, body 15-16, caption 12. Minimum tap target 48 px
Shape	Rounded corners 16 px, bottom sheet with drag handle, subtle shadows, generous spacing
Icons	Simple outline icons: store (open place), lamp (lighting), road, cross (help point), bus (transit), clock, info
Map	Mapbox GL dark style. Start and destination pins, route lines as colored segments (about 50 m each), help points as small cross markers, open places as small dots

6. User flow

Welcome (first time) -> Plan trip -> (optional) Settings -> Fetching live data -> Results: map + three route cards -> Route detail (reasons, gaps, AI explanation) -> optional Time comparison (8 PM vs 1 AM) -> optional Walk mode. Stretch: Heartbeat check setup and alert flow. Always reachable: About this data.

6b. Platforms and devices (design for all of these)

We ship **one responsive web app** (installable to the home screen) that must look and work the same on **Android (Chrome)** and **iPhone (Safari)**, plus an optional **Samsung Galaxy Watch (Wear OS)** companion for the heartbeat check. No separate native phone apps. Apple Watch is out of scope.

Topic	Design requirement
Screen sizes	Primary 375 x 812 (iPhone) and 360 x 800 (Android); also check 360 x 640 small phones. Desktop two-pane layout is a bonus
Safe areas	Keep content clear of the iPhone notch/Dynamic Island and home bar, and the Android gesture bar. Bottom sheet buttons sit above the home indicator
Keyboard	On-screen keyboard must not hide the From/To fields or the Show routes button; suggestions list scrolls above it
Controls	Use platform-neutral components; do not copy an iOS-only or Material-only look. Minimum 48 px targets; bottom sheet has a drag handle plus buttons (no gesture-only actions)
Install	Home-screen install tip: on iPhone it is Share -> Add to Home Screen; on Android a banner. Design a short "Install for best results" card
Orientation	Portrait is primary; landscape must not break (map plus collapsed sheet)
Dark/light	Dark default; light optional. Respect the phone setting
Limits to design around	iPhone Safari: no Bluetooth heart-rate band, no vibration, tracking pauses when the screen locks, push only after install. Android Chrome: Bluetooth band works. Always show a clear fallback message instead of a broken button
Watch	Round Wear OS screen about 450 x 450 px, glanceable, one action per screen, high contrast, haptic pattern for the prompt

7. Screens needed

Screens 1-7 and 10 are **must have** for the demo and must be designed for both Android and iPhone. Screens 8, 9 and 11 are **nice to have / stretch**. For each screen, design the default state plus the listed states.

Screen 1: Welcome / first-run *[Must]*

Purpose: Explain the idea in 10 seconds and set expectations honestly.

Elements:

- App name and one-line promise: "Walking routes for the time you will actually be on the street."
- Three short points with icons: three routes, reasons shown, time-aware
- Honesty card: "We describe conditions, we do not guarantee safety. In an emergency call 911."
- Button "Get started"; link "How scores work" (opens screen 10)
- Location permission explained in plain words: used only during your trip, deleted afterward, never shared unless you choose

States to design: default; location permission denied (offer manual start entry)

Screen 2: Plan trip (home) *[Must]*

Purpose: Enter where from/to and when. This is the main screen.

Elements:

- Full-screen dark map behind a bottom sheet
- From field (default "My location") and To field with search suggestions (geocoding); swap button between them
- Departure time chip: "Leave now" or a date and time picker. Helper text: "Scores use the time you reach each street, not when you leave"
- Quick chips for the demo: "Tonight 8 PM", "Tonight 1 AM"
- Primary button "Show routes"; secondary icon button "Settings"
- Small footer: "Live data covers the demo area: Palace of Fine Arts to Montgomery BART"

States to design: empty; typing with suggestions; outside demo area (friendly "We only have live data for the demo area"); location denied

Screen 3: Settings *[Must]*

Purpose: Let the user tune the result.

Elements:

- Maximum extra minutes: slider 0-20 min (default 10) with live label
- Walking alone / with someone: segmented control
- What matters most: chips (Lighting, Open places, Main roads, Help points, Transit); choose up to 2
- Reset to default; Done
- Note: "Past-incident data is used lightly because it can reflect policing as much as danger."

States to design: default; changed values; reset

Screen 4: Fetching live data *[Must]*

Purpose: Make the live data visible and keep users patient (also proof of live data for judges).

Elements:

- Progress checklist that fills in: "Finding routes", "Checking what's open when you arrive", "Checking lighting and main roads", "Checking transit", "Writing the explanation"
- Each line shows a plain source name (e.g. "Live business listings") and a tick, "slow" or "not available"
- Cancel button

States to design: in progress; slow source ("Still loading places, showing what we have"); one source failed (continues with a warning)

Screen 5: Results: map + three route cards *[Must]*

Purpose: The core screen: compare the three routes.

Elements:

- Map with the selected route drawn as colored segments; the other two faded; start and destination pins
- Bottom sheet (peek, half, full) with three route cards, tab-like selection
- Each card: label, total minutes and "+6 min vs fastest", route score with a short phrase ("Mostly lit and active"), coverage % (e.g. 86%), longest empty stretch (e.g. 120 m near the park), open places count, nearest help point and distance
- Merged case: one card with two badges and the note "Most lit & active and Balanced are the same route"
- "Data fetched at 10:31 PM" badge on the sheet
- Legend: segment colors + icons + "not sure" (dashed grey)
- Buttons: "Details" on each card, "Compare 8 PM vs 1 AM", "Start walk"

States to design: three distinct routes; two merged; no route within extra-time cap (show Fastest only, with explanation); some segments "not sure"

Screen 6: Route detail *[Must]*

Purpose: Show exactly why this route is rated the way it is.

Elements:

- Header: route label, minutes, score
- Continuity block (our key difference): coverage bar with an 80% target marker, longest empty stretch with location note, worst 200 m window score
- Segment strip: horizontal bar of the whole route colored by segment; tap a part to highlight it on the map
- Tapped segment popover: score, band, arrival time at that point, reasons such as "2 open places within 50 m", "lit main road", "no open places nearby"
- Factor breakdown, 8 small bars with weights: open places 25%, lighting 20%, main roads 15%, help points 10%, transit 10%, activity estimate 10%, past incidents 5%, turns and dead ends 5%; activity labeled "estimate"
- AI explanation card in plain language with "Explained by AI, may be wrong"
- Open places list (name, category, closes at, 24 h badge); data source and fetched-at time
- Buttons: "Walk this route", "Back"

States to design: full data; stale data ("older than 2 h" warning); segment confidence unknown

Screen 7: Time comparison: 8 PM vs 1 AM *[Must (demo moment)]*

Purpose: Show that scores change with the clock: shops close, coverage drops, the best route changes.

Elements:

- Two columns (or a swipe/toggle) for the same trip at 8 PM and 1 AM
- Each shows a map thumbnail with colored segments, coverage %, longest gap, open places count
- One-sentence change summary (sample): "At 1 AM coverage drops from 86% to 52% and the best route changes"
- Time slider to scrub between 6 PM and 3 AM, updating the numbers

States to design: same route both times; route changes; second time still loading

Screen 8: Walk mode *[Nice]*

Purpose: Simple guidance while walking.

Elements:

- Large next-turn instruction at top; map follows the user
- Upcoming stretch indicator: "Next 200 m: well lit and active" or "Next 150 m: quiet stretch"
- Small "I'm okay" and "End walk" buttons; a visible "Call 911" button
- Note: location is used only during the walk and deleted after

States to design: on route; off route (offer re-plan); arrived

Screen 9: Heartbeat check (opt-in) *[Stretch]*

Purpose: Optional alert to a trusted contact. Must feel optional, private and non-medical.

Elements:

- 9a Consent: what is read, why, stays on the phone, deleted after the trip; toggle off by default
- 9b Setup: choose contact (name and phone); the contact must agree to receive alerts (status pending/agreed); sensitivity setting
- 9c Device choice: "Pair my Samsung watch" (shows a 6-digit pairing code to type on the watch, then live bpm), "Connect Bluetooth band" (Android Chrome only; disabled with explanation on iPhone), or "Use simulated reading" with a permanent **Simulated** badge
- 9d Check-in prompt: "Unusual heart rate detected. Are you okay?" with a big "I'm okay" button and a 30-second countdown ring
- 9e Alert sent: confirms the contact was messaged with live location; "Call 911" button; "Cancel alert" if sent in error
- Footer on all: "Does not replace calling 911."

States to design: consent off/on; contact pending; countdown running; alert sent; cancelled; input not supported on this browser (iPhone): show "Heart-rate input is not supported on this browser. Use a paired watch or the simulated reading."

Screen 10: About this data and how scores work *[Must (small)]*

Purpose: Responsible design and judge credibility.

Elements:

- Plain explanation of the 8 factors and weights, and the continuity rule with a tiny diagram (coverage, longest gap, worst window)
- Data sources list with what each provides and last-updated time (live business listings, OpenStreetMap, 311 outage reports, transit schedules, public incident data)
- Honest limits: activity is an estimate; incident data has low weight and can unfairly flag areas; data can be missing; AI explanations can be wrong
- Privacy: minimal location, deleted after the trip, sharing is opt-in
- Emergency: call 911

States to design: default

Screen 11: Samsung Wear OS watch app *[Stretch]*

Purpose: Glanceable companion that reads heart rate and asks "Are you OK?" even when the phone is locked.

Elements:

- 11a Pair: enter the 6-digit code shown on the phone (big number pad), "Paired" confirmation
- 11b Walking: small status "Walking - heart check on", live bpm with a Simulated badge when simulated, optional next-stretch hint ("Quiet stretch in 150 m"), "End walk"
- 11c Prompt: "Are you OK?" with a big "I'm OK" button, 30-second countdown ring, haptic buzz
- 11d Alert sent: "Contact notified", "Cancel alert", small "Call 911"
- 11e Permission needed: one-line explanation before asking for heart-rate access

States to design: paired; unpaired; prompt counting down; alert sent; cancelled; no heart-rate permission; phone disconnected

8. Reusable components to define

- Route card (3 variants + merged variant), segment legend, coverage bar with target marker, segment strip, factor bar, reason chip, data-fetched-at badge, "not sure" tag, "estimate" tag, "simulated" badge, bottom sheet (peek/half/full), time picker chip, source-status row, install card, "not supported on this browser" notice, watch pairing keypad, watch countdown ring, warning banner (slow/stale/blocked), empty and error placeholders, primary/secondary/destructive buttons

9. Data each screen can display (nothing else exists)

Item	Fields
Route	label, also_labels, duration_min, extra_min, distance_m, route_score, open_places_count, nearest_help_point{name, distance_m}, explanation, explanation_source (ai or template)
Continuity	coverage_pct, longest_gap_m, longest_gap_note, worst_window_score
Segment	segment_id, start, end, length_m, arrival_time, score, band (green/yellow/red/unknown), factors, reasons[], data_fetched_at, confidence (high/low/unknown)
Response	api_version, scoring_version, data_fetched_at, warnings[]
Not available	Live people counts, crime predictions, a guaranteed safety rating. Do not design UI for these.

10. Sample content for mockups

- Trip: Palace of Fine Arts to Montgomery BART, leaving 10:40 PM, about 3.5 km.
- Most lit & active: 46 min (+5), score 78, coverage 88%, longest gap 110 m near the Presidio edge, 14 open places, nearest help point 90 m.
- Balanced: 43 min (+2), score 66, coverage 74%, longest gap 220 m near the park, 9 open places.
- Fastest: 41 min, score 49, coverage 52%, longest gap 380 m, 5 open places.
- Reason examples: "2 open places within 50 m", "lit main road", "bus stop with service in the next 10 min", "no open places for 220 m", "not sure: lighting data missing".
- Mark all these numbers as sample in the files you hand over.

11. Deliverables requested from Stitch

- Mobile screens 1-7 and 10 (and 8, 9, 11 if time allows) in the dark theme, with the listed states, at iPhone and Android sizes (including a small-phone check of screen 5).
- One desktop layout of screen 5 (results) for the demo projector.
- A one-page component sheet (colors, type, route card, legend, badges).
- Exportable code or images developers can use. The phone frontend is a Next.js responsive web app with Mapbox GL JS, so keep the map as a full-bleed layer with overlays on top. The watch app is native Wear OS (Kotlin); provide watch screens as images with sizes and colors.

12. Paste-ready prompt for Stitch

Design a mobile-first, dark-theme web app called Safe Smart Maps. It compares three walking routes (Most lit & active, Balanced, Fastest) for a late-night student, scoring each on open places, lighting, main roads, help points and transit, and on continuity along the whole route (coverage %, longest empty stretch, worst 200 m window). Never use the word safe as a promise; use better lit and more active, and always show reasons, a data-fetched-at time, and a not sure state. Screens: Welcome, Plan trip (from/to, departure time, quick chips 8 PM and 1 AM), Settings (extra minutes slider, alone or with someone, priorities), Fetching live data checklist, Results (full-bleed map with segments colored green/yellow/red plus icons and patterns, bottom sheet with three route cards), Route detail (continuity block, segment strip, factor bars, AI explanation, open places), Time comparison 8 PM vs 1 AM, About this data. Must work on Android and iPhone browsers (safe areas, keyboard, small phones). Stretch: Walk mode, an opt-in heartbeat check flow (consent, contact, device choice, check-in countdown, alert sent) with a visible Simulated badge, and Samsung Wear OS watch screens (pair code, walking status, Are you OK prompt with countdown, alert sent). Calm tone, large tap targets, accessible contrast, color never the only signal.

13. Constraints and notes

- Scope is one demo area, not a whole city; do not design city pickers.
- No accounts or login screens. No payment screens.
- Do not rely on hover, long-press or gesture-only actions; no feature that exists on only one platform without a fallback.
- Apple Watch and separate native phone apps are out of scope.
- Anything not listed in section 9 does not exist in the data; ask before adding it.
- Questions or contract changes go to Bhavpreet or Amey. Shared files: docs/agent-context/CONTRACTS.md and docs/specs/FUNCTIONAL_SPEC.md.